Vaishnavi Singla

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Education

Thapar Institute of Engineering and Technology — **July 2024 – July 2028**
Bachelor of Engineering in Computer Science & Engineering ***(CGPA: 8.37)*** — *Patiala, Punjab*

Experience

ELC Summer Intern — **June 2026 – July 2026**
Thapar Institute of Engineering & Technology — *Patiala, Punjab*

- Developed a **Multimodal AI Framework for Pregnancy Risk Assessment** using maternal clinical data and fetal ultrasound images.
- Evaluated **Gradient Boosting** on clinical data, and **transfer learning-based ResNet-18** initialized with **pretrained ImageNet weights** for fetal ultrasound classification, both using **5-fold cross-validation**.
- Gradient Boosting achieved an **accuracy of 99.01%, F1-score of 99.01%**, while ResNet-18 achieved an **accuracy of 92.27%, F1-score of 95.4%**.
- Designed an **embedding-based late-fusion framework** integrating clinical predictions and ultrasound image analysis for pregnancy risk assessment achieving **98.8% accuracy and 98.5% F1-score**.
- GitHub: https://github.com/vaishnavisingla222/PregnancyrRiskELC
- Live: https://pregnancy-predictor-835v.vercel.app/

Technical Skills

Languages: C, C++, Java, Python
Libraries and Frameworks: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow, Keras, Streamlit
Databases: MySQL, Oracle SQL, PL/SQL
Web Technologies: CSS, HTML, Java Script
Tools: Git, GitHub, VS Code, Jupyter Notebook
CS Fundamentals: Data Structures and Algorithms, Operating Systems, Database Management System, Object Oriented Programming, Artificial Intelligence, Machine Learning, Probability & Statistics, Computer Networks

Projects

Digital Payment Fraud Detection | *Oracle SQL, PL/SQL* — **April - May 2026**

- Developed PL/SQL modules including **1 trigger, 1 procedure, 1 function, and 1 cursor** to automate **fraud detection, transaction processing, fraud logging, and account blocking**.
- Supported **multiple payment modes including UPI, wallet, card, and net banking** with rule-based anomaly detection.
- Wrote **50+ SQL queries** to analyze fraud patterns, user behavior, and transaction trends.
- GitHub: https://github.com/vaishnavisingla222/Digital_Payment_Fraud_Detection

Loan Approval Prediction System | *Python, Numpy, Pandas, Matplotlib/Seaborn, Scikit-learn, Streamlit* — **April 2026**

- Engineered a **supervised machine learning** pipeline to predict loan approval using **4,269 records with 13 features** including 1 targeted feature.
- Compared **Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, and XGBoost**, selecting **Random Forest** as the final deployment model.
- Achieved **97.78% accuracy, 97.78% precision, 98.69% recall, and 98.24% F1-score** through hyperparameter tuning and class imbalance handling.
- GitHub: https://github.com/vaishnavisingla222/loan_approval_ml
- Live: https://loanapprovalpridiction-h25gkuw8flzcb8myaubv8q.streamlit.app/

Achievements

- **400+ DSA problems** solved on **LeetCode**, **GeeksforGeeks**, and **CodeChef** focusing on efficient and readable solutions using **Java**.
- **Merit III Scholarship** awarded for exceptional academic performance, amounting to **about 3 Lakhs**.